

7. The number of parallel paths in the armature = mP where m is the multiplicity of the winding and P the number of poles.

Taking the first condition, we have $Y_B = Y_F \pm 2$.**

(a) If $Y_B > Y_F$ i.e. $Y_B = Y_F + 2$, then we get a progressive or right-handed winding i.e. a winding which progresses in the clockwise direction as seen from the commutator end. In this case, obviously, $Y_C = +1$.

(b) If $Y_B < Y_F$ i.e. $Y_B = Y_F - 2$, then we get a retrogressive or left-handed winding i.e. one which advances in the anti-clockwise direction when seen from the commutator side. In this case, $Y_C = -1$.

(c) Hence, it is obvious that

$$\left. \begin{aligned} Y_F &= \frac{Z}{P} - 1 \\ Y_B &= \frac{Z}{P} + 1 \end{aligned} \right\} \text{ for progressive winding and } \left. \begin{aligned} Y_F &= \frac{Z}{P} + 1 \\ Y_B &= \frac{Z}{P} - 1 \end{aligned} \right\} \text{ for retrogressive winding}$$

Obviously, Z/P must be even to make the winding possible.

26.27. Numbering of Coils and Commutator Segments

In the d.c. winding diagrams to follow, we will number the coils only (not individual turns). The upper side of the coil will be shown by a firm continuous line whereas the lower side will be shown by a broken line. The numbering of coil sides will be consecutive i.e. 1, 2, 3 etc. and such that odd numbers are assigned to the top conductors and even numbers to the lower sides for a two-layer winding. The commutator segments will also be numbered consecutively, the number of the segments will be the same as that of the upper side connected to it.

Example 26.1. Draw a developed diagram of a simple 2-layer lap-winding for a 4-pole generator with 16 coils. Hence, point out the characteristics of a lap-winding.

(Elect. Engineering, Madras Univ. 1981)

Solution. The number of commutator segments = 16

Number of conductors or coil sides $16 \times 2 = 32$; pole pitch = $32/4 = 8$

Now remembering that (i) Y_B and Y_F have to be odd and (ii) have to differ by 2, we get for a progressive winding $Y_B = 9$; $Y_F = -7$ (retrogressive winding will result if $Y_B = 7$ and $Y_F = -9$). Obviously, commutator pitch $Y_C = -1$.

[Otherwise, as shown in Art. 26.26, for progressive winding

$$Y_F = \frac{Z}{P} - 1 = \frac{32}{4} - 1 = 7 \text{ and } Y_B = \frac{Z}{P} - 1 = \frac{32}{4} + 1 = 9]$$

The simple winding table is given as under :

Back Connections

1 to $(1 + 9) = 10$
3 to $(3 + 9) = 12$
5 to $(5 + 9) = 14$
7 to $(7 + 9) = 16$
9 to $(9 + 9) = 18$
11 to $(11 + 9) = 20$
13 to $(13 + 9) = 22$
15 to $(15 + 9) = 24$
17 to $(17 + 9) = 26$
19 to $(19 + 9) = 28$

Front Connections

————→ 10 to $(10 - 7) = 3$
————→ 12 to $(12 - 7) = 5$
————→ 14 to $(14 - 7) = 7$
————→ 16 to $(16 - 7) = 9$
————→ 18 to $(18 - 7) = 11$
————→ 20 to $(20 - 7) = 13$
————→ 22 to $(22 - 7) = 15$
————→ 24 to $(24 - 7) = 17$
————→ 26 to $(26 - 7) = 19$
————→ 28 to $(28 - 7) = 21$

** In general, $Y_B = Y_F \pm 2m$ where $m = 1$ for simplex lap winding and $m = 2$ for duplex lap winding etc.

$$21 \text{ to } (21 + 9) = 30$$

$$23 \text{ to } (23 + 9) = 32$$

$$25 \text{ to } (25 + 9) = 34 = (34 - 32) = 2$$

$$27 \text{ to } (27 + 9) = 36 = (36 - 32) = 4$$

$$29 \text{ to } (29 + 9) = 38 = (38 - 32) = 6$$

$$31 \text{ to } (31 + 9) = 40 = (40 - 32) = 8$$

$$30 \text{ to } (20 - 7) = 23$$

$$32 \text{ to } (32 - 7) = 25$$

$$2 \text{ to } (34 - 7) = 27$$

$$4 \text{ to } (36 - 7) = 29$$

$$6 \text{ to } (38 - 7) = 31$$

$$8 \text{ to } (40 - 7) = 33 = (33 - 32) = 1$$

The winding ends here because we come back to the conductor from where we started.

We will now discuss the developed diagram which is one that is obtained by imagining the armature surface to be removed and then laid out flat so that the slots and conductors can be viewed without the necessity of turning round the armature in order to trace out the armature windings. Such a developed diagram is shown in Fig. 26.31.

